

SEBASTIÁN HERMOSILLA CANOBRA

Master of Science in Climate Physics

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STUDIES

Master of Science in Climate Physics

Utrecht University

 September 2024 – July 2026  Utrecht, Netherlands

- Awarded the “NL Scholarship” for academic excellence

Bachelor of Science in Astronomy

University of Chile

 March 2019 – August 2023  Santiago, Chile

RESEARCH EXPERIENCE

Master Thesis: Circulation and cloud-cover fingerprints in aquaplanet atmospheres

Utrecht University

 November 2025 – July 2026  Utrecht, Netherlands

- Investigating circulation responses to different planetary rotation rates and obliquities. The main atmospheric structures were mapped in the studied parameter space and distinct cloud cover responses were found for different rotation rates at high obliquities. Main supervisor: Prof. Henk Dijkstra

Glacier Monitoring with Satellites and Seismic Waves

Data Observatory

 February – May 2023  Santiago, Chile

- Led the winning team in the national “DO-safios” competition, developing a methodology combining satellite imagery and seismic data to quantify glacier retreat and detect large calving events at Grey Glacier. Competition: <https://www.dosafios.cl/>

Asteroids Research Project

Department of Astronomy, U. Chile

 August 2023-August 2024  Santiago, Chile

- Conducted observational campaigns during twilight periods to detect and characterize near-Earth asteroids. Contributing to survey design and data analysis. Supervisor: Prof. Patricio Rojo.

Planetary Atmospheres Research Project

Department of Astronomy, U. Chile

 August 2022-August 2023  Santiago, Chile

- Developed a catalog with the atmospheric properties of various solar system bodies to compare them with future observations of exoplanets using VLT observations. Tutor: Prof. Patricio Rojo.
- Publication: “HiDef Neighbors: Solar System Objects as Exoplanet Analogues” A&A, 707 (2026) A325
doi: <https://doi.org/10.1051/0004-6361/202555769>

CAREER GOALS

“Build knowledge through observation, research and teaching.”

ACHIEVEMENTS



FCFM Outstanding Student (3x)

Selected as an outstanding student of the Faculty of Physical and Mathematical Sciences of University of Chile in 2020, 2021 and 2022.



National PSU score

2018 - Obtained the highest score in the Mathematics test for university admission, along with the highest score in Los Lagos region (Chile) in the Science test.



University Cross Country and Athletics Team

2023 - Third place in the National Championship for Universities as a member of the Cross Country team.

National Champion

2015 - National Champion as a member of the Puerto Montt city basketball team.

STRENGTHS

Creativity

Proactivity

Teamwork

Responsibility

Solidarity

SKILLS

Data analysis & visualization

Numerical problem solving

Physics and mathematics

Machine learning

LANGUAGES

Spanish (Native)

English (Advanced)

German (Basic)



OBSERVATIONAL EXPERIENCES

Cerro Tololo Inter-American Observatory
Association of Universities for Research in Astronomy (AURA)

June - Sep. 2023 Cerro Tololo, Chile

- **4-meter Víctor M. Blanco telescope:** Carried out a 5-night twilight campaign to identify asteroids within Earth's orbit, totaling around 5 hours of observation.
- **0.9-meter SMARTS Telescope:** Conducting planetary transit observations for a total of ~80 hours, split across two periods of 4 and 6 nights each.

TEACHING EXPERIENCE

Planetary Science teaching assistant
Department of Astronomy, U. Chile

March – August 2024 Santiago, Chile

- Responsible for conducting one class per week, preparing and grading tests. Head teacher: Patricio Rojo

Introduction to Classical Physics teaching assistant
Department of Physics, U. Chile

March – August 2024 Santiago, Chile

- Responsible for conducting one class per week, grading tests and conducting laboratory activities. Head teacher: Valentino González.

OTHER EXPERIENCES

Students Representative
University of Utrecht

September 2024-Present Utrecht, Netherlands

- Responsible for serving as the communication channel between the professors and students of the master's program, attending meetings with the professors on behalf of the students.

Postgraduate Symposium
University of Chile

August 2024 Santiago, Chile

- Presented a poster at the Postgraduate Symposium and won the award for the best 1-minute thesis presentation.

OTHER INTERESTS

Apart from science, I have a range of other interests:

- **University Track and Field Team**
Specialized in 10,000m and 5,000m races, between 2022 and 2024.
- **Engineering and Sciences Diffusion Group**
Led efforts to bridge university research with Chilean society and promote faculty enrollment among school students. between 2022-2023.
- **TECHO for Chile Foundation Member**
Volunteer in the construction of housing for communities in crisis in 2019 and 2020.
- **Basic Mountaineering Course**
Certified in Basic Mountaineering by the University Andean Club of Chile in 2020, gaining skills in navigation, displacement techniques and first aid.

REFERENCES

Prof. Henk Dijkstra
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Prof. Patricio Rojo
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Department of Astronomy, U Chile

Prof. Valentino González
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